Yarim va chorak doira yuzi

Yarim doira shaklidagi gulzorga qancha maysa ekiladi? Chorak doira shaklidagi burchak javoni ustiga qancha laminat kerak? Buni bilish uchun yarim va chorak doiraning yuzini topamiz.

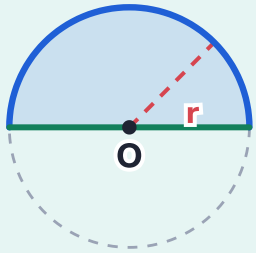
1-QISM

Mavzu

USUL · YARIM DOIRA YUZI

$$\text{Radius} = \text{diametr} : 2$$

$$\text{Yarim doira yuzi} = \frac{1}{2} \times \pi r^2$$



1-MISOL · DIAMETR BERILGAN

Yarim doiraning diametri 8,4 dm. Uning yuzini toping ($\pi = \frac{22}{7}$).



1. Radius: $8{,}4 : 2 = 4{,}2$ dm.
2. $\frac{1}{2} \cdot \frac{22}{7} \cdot 4{,}2 \cdot 4{,}2$. Maxrajdagi 7 qisqaradi: $4{,}2 : 7 = 0{,}6$.
3. $\frac{1}{2} \cdot 22 \cdot 0{,}6 \cdot 4{,}2 = 11 \cdot 0{,}6 \cdot 4{,}2 = 27{,}72\ \text{dm}^2$.

2-MISOL · RADIUS BERILGAN

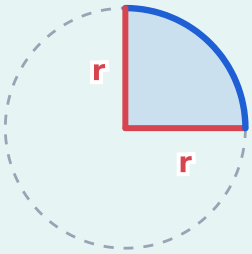
Yarim doiraning radiusi 6 cm. Uning yuzini toping ($\pi = 3{,}14$).



1. $r^2 = 6 \cdot 6 = 36$.
2. Yuzi: $\frac{1}{2} \cdot 3{,}14 \cdot 36 = 1{,}57 \cdot 36 = 56{,}52\ \text{cm}^2$.

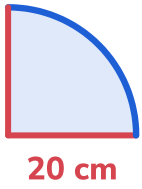
USUL · CHORAK DOIRA YUZI

$$\text{Chorak doiraning yuzi} = \frac{1}{4} \times \pi r^2$$



1-MISOL · RADIUS BERILGAN

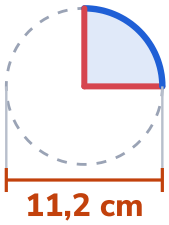
Chorak doiraning radiusi 20 cm. Uning yuzini toping ($\pi = 3{,}14$).



1. $r^2 = 20 \cdot 20 = 400$.
2. Yuzi: $\frac{1}{4} \cdot 3{,}14 \cdot 400 = 3{,}14 \cdot 100 = 314$ cm^2.

2-MISOL · DIAMETR BERILGAN

Doiraning diametri 11,2 cm. Uning choragi yuzini toping ($\pi = \frac{22}{7}$).



1. Radius: $11{,}2 : 2 = 5{,}6$ cm.
2. $\frac{1}{4} \cdot \frac{22}{7} \cdot 5{,}6 \cdot 5{,}6$. Maxrajdagi 7 qisqaradi: $5{,}6 : 7 = 0{,}8$.
3. $0{,}8 : 4 = 0{,}2$. Yuzi: $22 \cdot 0{,}2 \cdot 5{,}6 = 24{,}64$ cm^2.

Yarim doirada doira yuzini 2 ga, chorak doirada 4 ga bo'lamiz. Radius esa o'zgarmaydi — uni ikkiga bo'lmang.

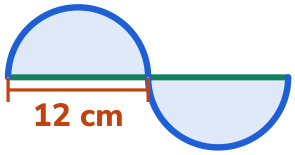


USUL · BO'LAKLARDAN TUZILGAN SHAKL

1. Shakl qanday bo'laklardan tuzilganini aniqlang: yarim doiralar, chorak doiralar.
2. Har bir bo'lakning yuzini toping.
3. Yuzlarni qo'shing. Bo'laklar birga butun doira hosil qilsa, darhol πr^2 ni toping.

1-MISOL · IKKITA YARIM DOIRA

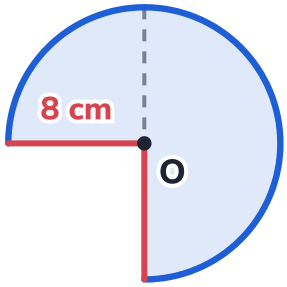
Shakl ikkita bir xil yarim doiradan tuzilgan. Har birining diametri 12 cm. Shaklning yuzini toping ($\pi = 3{,}14$).



1. Ikkita bir xil yarim doira birga — bitta butun doira. Radius: $12 : 2 = 6$ cm.
2. Yuzi: $3{,}14 \cdot 6 \cdot 6 = 113{,}04$ cm^2.

2-MISOL · YARIM VA CHORAK DOIRA

Shakl yarim doira va chorak doiradan tuzilgan, markazi O, radiusi 8 cm. Shaklning yuzini toping ($\pi = 3{,}14$).



1. Yarim va chorak birga: $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ — doiraning $\frac{3}{4}$ qismi.
2. Doira: $3{,}14 \cdot 8 \cdot 8 = 200{,}96$ cm^2. Uning $\frac{3}{4}$ qismi: $200{,}96 : 4 \cdot 3 = 150{,}72$ cm^2.



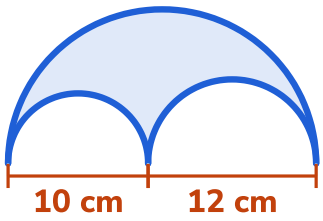
Radiuslari teng bo'lsa, 2 ta yarim doira yoki 4 ta chorak doira birga bitta butun doira bo'ladi.

USUL · BO'YALGAN SOHANING YUZI

1. Katta shaklning yuzini toping.
2. Kesib olingan (bo'yalmagan) bo'laklarning yuzini toping.
3. Katta shakl yuzidan kesib olingan bo'laklar yuzini ayiring.

1-MISOL · UCHTA YARIM DOIRA

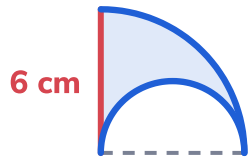
Diametri 22 cm bo'lgan yarim doiradan diametrlari 10 cm va 12 cm bo'lgan ikkita yarim doira kesib olindi. Bo'yalgan sohaning yuzini toping ($\pi = 3{,}14$).



1. Katta yarim doira, $r = 11$ cm: $\frac{1}{2} \cdot 3{,}14 \cdot 11 \cdot 11 = 189{,}97\ \text{cm}^2$.
2. Kichiklari, $r = 5$ cm va $r = 6$ cm: $\frac{1}{2} \cdot 3{,}14 \cdot 5 \cdot 5 = 39{,}25\ \text{cm}^2$ va $\frac{1}{2} \cdot 3{,}14 \cdot 6 \cdot 6 = 56{,}52\ \text{cm}^2$.
3. Bo'yalgan soha: $189{,}97 - 39{,}25 - 56{,}52 = 94{,}2\ \text{cm}^2$.

2-MISOL · CHORAK DOIRADAN YARIM DOIRA

Radiusi 6 cm bo'lgan chorak doiradan diametri 6 cm bo'lgan yarim doira kesib olindi. Qolgan qismning yuzini toping ($\pi = 3{,}14$).



1. Chorak doira: $\frac{1}{4} \cdot 3{,}14 \cdot 6 \cdot 6 = 28{,}26\ \text{cm}^2$.
2. Yarim doira, $r = 6 : 2 = 3$ cm: $\frac{1}{2} \cdot 3{,}14 \cdot 3 \cdot 3 = 14{,}13\ \text{cm}^2$.
3. Qolgan qism: $28{,}26 - 14{,}13 = 14{,}13\ \text{cm}^2$.

2-QISM

Misollar ishlash

1-MISOL · KALKULYATORDAGI π, YAXLITLASH

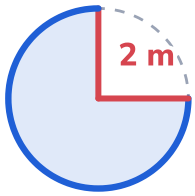
Yarim doira shaklidagi derazaning diametri 1,6 m. Uning oynasi yuzini kalkulyatordagi π tugmasi bilan toping. Javobni yuzdan birlar xonasigacha yaxlitlang.

1. Radius: $1{,}6 : 2 = 0{,}8$ m.
2. Kalkulyatorda: $\pi \times 0{,}8 \times 0{,}8 \div 2$. Ekranda $1{,}0053\ldots$ chiqadi.
3. Verguldan keyingi uchinchi raqam 5: $1{,}0053\ldots \approx 1{,}01\ \text{m}^2$.

2-MISOL · DOIRANING $\frac{3}{4}$ QISMI

Doiradan uning choragi kesib olindi. Doiraning radiusi 2 m. Qolgan shaklning yuzini toping ($\pi = 3{,}14$).

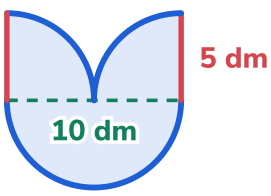
1. Qolgan shakl — doiraning $\frac{3}{4}$ qismi. Doira: $3{,}14 \cdot 2 \cdot 2 = 12{,}56\ \text{m}^2$, $\frac{3}{4}$ qismi: $12{,}56 : 4 \cdot 3 = 9{,}42\ \text{m}^2$.
2. Ikkinchi yo'l — ayirish: doira $3{,}14 \cdot 2 \cdot 2 = 12{,}56\ \text{m}^2$, chorak $12{,}56 : 4 = 3{,}14\ \text{m}^2$. $12{,}56 - 3{,}14 = 9{,}42\ \text{m}^2$.



3-MISOL · YARIM DOIRA VA IKKITA CHORAK DOIRA

Lola gulining rasmi diametri 10 dm bo'lgan yarim doira va radiusi 5 dm bo'lgan ikkita chorak doiradan tuzilgan. Shaklning yuzini toping ($\pi = 3{,}14$).

1. Yarim doiraning radiusi: $10 : 2 = 5$ dm. Chorak doiralarniki ham 5 dm — radiuslar teng.
2. Ulushlar: $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$ — bo'laklar birga bitta butun doiraga teng.
3. Yuzi: $3{,}14 \cdot 5 \cdot 5 = 78{,}5\ \text{dm}^2$.

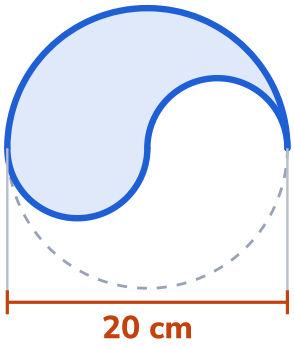


Radiuslar har xil bo'lsa, bo'laklar butun doira hosil qilmaydi. Unda har bir bo'lakning yuzi alohida topilib, qo'shiladi.

4-MISOL · BIR BO'LAK QO'SHILDI, XUDDI SHUNDAY BO'LAK OLINDI

Doiraning diametri 20 cm. Ichidagi egri chiziq ikkita bir xil yarim aylanadan iborat. Bo'yalgan sohaning yuzini toping ($\pi = 3{,}14$).

1. Bo'yalgan soha = katta yarim doira + chapdagi kichik yarim doira − o'ngdagi kichik yarim doira.
2. Kichik yarim doiralar bir xil: qo'shilgani olinganiga teng. Demak, bo'yalgan soha katta yarim doiraga teng.
3. $r = 20 : 2 = 10$ cm. $\frac{1}{2} \cdot 3{,}14 \cdot 10 \cdot 10 = 157\ \text{cm}^2$.





Yuzga diametr qo'shilmaydi! Diametr perimetrga qo'shilardi.

DIQQAT!

1. Diametr berilsa, avval radiusni toping. Chorak doira, $d = 20$ cm: ✗ $\frac{1}{4} \cdot 3{,}14 \cdot 20 \cdot 20$ ✓ $\frac{1}{4} \cdot 3{,}14 \cdot 10 \cdot 10 = 78{,}5\ \text{cm}^2$
2. $r^2 = r \cdot r$. Yarim doira, $r = 5$ cm: ✗ $\frac{1}{2} \cdot 3{,}14 \cdot 5 \cdot 2$ ✓ $\frac{1}{2} \cdot 3{,}14 \cdot 5 \cdot 5 = 39{,}25\ \text{cm}^2$
3. Yuzga diametr yoki radius qo'shilmaydi. Yarim doira, $r = 6$ cm: ✗ $56{,}52 + 12$ ✓ $56{,}52\ \text{cm}^2$
4. Yarim doirada yuz ikkiga bo'linadi, radius emas. $r = 10$ cm: ✗ $3{,}14 \cdot 5 \cdot 5$ ✓ $\frac{1}{2} \cdot 3{,}14 \cdot 10 \cdot 10 = 157\ \text{cm}^2$
5. Yuz kvadrat birlikda yoziladi: ✗ 157 cm ✓ $157\ \text{cm}^2$

3-QISM

Mustahkamlash

1. Bo'sh joyni to'ldiring.
   a) Yarim doira yuzi $= \ldots \times \pi r^2$
   b) Chorak doiraning yuzi $= \ldots \times \pi r^2$
   c) Radiusi bir xil 2 ta yarim doira birga ... hosil qiladi.
   d) Radiusi bir xil yarim doira va chorak doira birga doiraning ... qismini hosil qiladi.

2. Yarim doira yuzini toping ($\pi = 3{,}14$).
   a) $r = 4$ cm
   b) $r = 20$ m
   c) $d = 6$ dm
   d) $d = 1$ m.

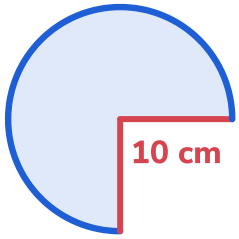
3. Yarim doira yuzini toping ($\pi = \frac{22}{7}$).
   a) $r = 3{,}5$ cm
   b) $d = 5{,}6$ m
   c) $r = 63$ mm.

4. Chorak doira yuzini toping ($\pi = 3{,}14$).
   a) $r = 2$ m
   b) $r = 30$ cm
   c) $r = 1{,}2$ dm
   d) butun doiraning diametri 10 cm.

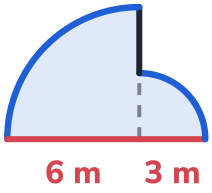
5. Chorak doira yuzini toping ($\pi = \frac{22}{7}$).
   a) $r = 0{,}7$ m
   b) $r = 4{,}2$ cm
   c) $r = 70$ mm.

6. Yuzni kalkulyatordagi π tugmasi bilan toping va yuzdan birlar xonasigacha yaxlitlang.
   a) Yarim doira, $d = 11$ cm.
   b) Chorak doira, $r = 2{,}5$ m.

7. Shakllarning yuzini toping ($\pi = 3{,}14$).

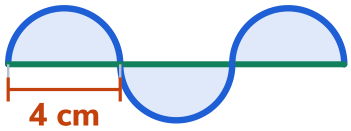


a) doiraning $\frac{3}{4}$ qismi

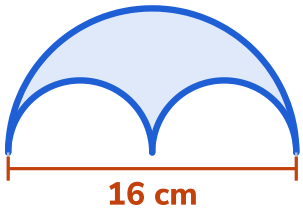


b) ikkita chorak doira

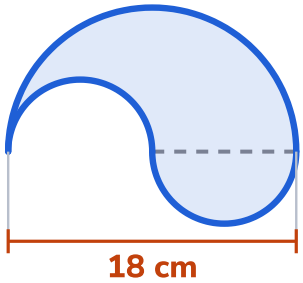
8. Shakl 3 ta bir xil yarim doiradan tuzilgan. Uning yuzini toping ($\pi = 3{,}14$).



9. Diametri 16 cm bo'lgan yarim doiradan ikkita bir xil kichik yarim doira kesib olindi. Bo'yalgan sohaning yuzini toping ($\pi = 3{,}14$).



10. Katta yarim doiraning diametri 18 cm. Diametrning chap yarmiga tayangan kichik yarim doira kesib olindi, o'ng yarmiga esa xuddi shunday yarim doira pastdan yopishtirildi. Shaklning yuzini toping ($\pi = 3{,}14$).



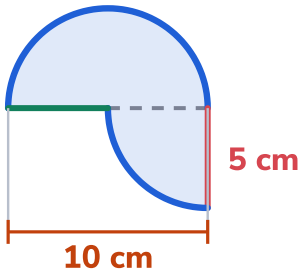
11. Hovuz yonidagi chorak doira shaklidagi maydonchaga plitka yotqiziladi. Maydonchaning radiusi 4 m. Har $1\ \text{m}^2$ ga 25 ta plitka ketadi. Nechta plitka kerak? ($\pi = 3{,}14$)

12. Radiusi 16 cm bo'lgan dumaloq pishiriq 4 ta teng bo'lakka kesildi. Bitta bo'lakning ustki yuzi qancha? ($\pi = 3{,}14$)

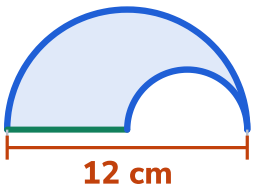
4-QISM

Mustaqil ish

1. Diametri 26 cm bo'lgan yarim doiraning yuzini toping ($\pi = 3{,}14$).
2. Radiusi 2,1 m bo'lgan chorak doiraning yuzini toping ($\pi = \frac{22}{7}$).
3. Yarim doiraning diametri 7,4 cm. Uning yuzini kalkulyatordagi π tugmasi bilan toping va o'ndan birlar xonasigacha yaxlitlang.
4. Doiraning $\frac{3}{4}$ qismidan iborat shaklning radiusi 2,8 dm. Shaklning yuzini toping ($\pi = \frac{22}{7}$).
5. Shakl diametri 10 cm bo'lgan yarim doira va radiusi 5 cm bo'lgan chorak doiradan tuzilgan. Uning yuzini toping ($\pi = 3{,}14$).



6. Diametri 12 cm bo'lgan yarim doiradan diametri 6 cm bo'lgan yarim doira kesib olindi. Qolgan qismning yuzini toping ($\pi = 3{,}14$).



7. Yarim doira shaklidagi ayvonning poli bo'yaladi. Ayvonning diametri 6 m. Har $1\ \text{m}^2$ ga 150 g bo'yoq ketadi. 2 kg bo'yoq yetadimi? ($\pi = 3{,}14$)
8. **Xatoni toping.** Nodira radiusi 12 cm bo'lgan chorak doiraning yuzini shunday topdi: $\frac{1}{2} \cdot 3{,}14 \cdot 12 \cdot 12 = 226{,}08\ \text{cm}^2$. Xato nimada? To'g'ri javob qancha?
9. **Qiyinroq.** Chorak doiraning radiusi 10 cm, yarim doiraning diametri 10 cm ($\pi = 3{,}14$).
   a) Ikkala shaklning yuzini toping.
   b) Qaysi birining yuzi katta va necha marta?

5-QISM

Javoblar

Mustahkamlash

1-masala

a) $\frac{1}{2}$; b) $\frac{1}{4}$; c) butun doira; d) $\frac{3}{4}$ qismini.

2-masala

a) $\frac{1}{2} \cdot 3{,}14 \cdot 16 = 25{,}12\ \text{cm}^2$; b) $\frac{1}{2} \cdot 3{,}14 \cdot 400 = 628\ \text{m}^2$; c) $r = 3$ dm, $\frac{1}{2} \cdot 3{,}14 \cdot 9 = 14{,}13\ \text{dm}^2$; d) $r = 0{,}5$ m, $\frac{1}{2} \cdot 3{,}14 \cdot 0{,}25 = 0{,}3925\ \text{m}^2$.

3-masala

a) $\frac{1}{2} \cdot 22 \cdot 0{,}5 \cdot 3{,}5 = 19{,}25\ \text{cm}^2$; b) $r = 2{,}8$ m, $\frac{1}{2} \cdot 22 \cdot 0{,}4 \cdot 2{,}8 = 12{,}32\ \text{m}^2$; c) $\frac{1}{2} \cdot 22 \cdot 9 \cdot 63 = 6237\ \text{mm}^2$.

4-masala

a) $\frac{1}{4} \cdot 3{,}14 \cdot 2 \cdot 2 = 3{,}14\ \text{m}^2$; b) $\frac{1}{4} \cdot 3{,}14 \cdot 900 = 706{,}5\ \text{cm}^2$; c) $\frac{1}{4} \cdot 3{,}14 \cdot 1{,}44 = 1{,}1304\ \text{dm}^2$; d) $r = 5$ cm, $\frac{1}{4} \cdot 3{,}14 \cdot 25 = 19{,}625\ \text{cm}^2$.

5-masala

a) $\frac{1}{4} \cdot 22 \cdot 0{,}1 \cdot 0{,}7 = 0{,}385\ \text{m}^2$; b) $\frac{1}{4} \cdot 22 \cdot 0{,}6 \cdot 4{,}2 = 13{,}86\ \text{cm}^2$; c) $\frac{1}{4} \cdot 22 \cdot 10 \cdot 70 = 3850\ \text{mm}^2$.

6-masala

a) $r = 5{,}5$ cm. $\pi \times 5{,}5 \times 5{,}5 \div 2 = 47{,}516\ldots \approx 47{,}52\ \text{cm}^2$. b) $\pi \times 2{,}5 \times 2{,}5 \div 4 = 4{,}908\ldots \approx 4{,}91\ \text{m}^2$.

7-masala

a) $\frac{3}{4} \cdot 3{,}14 \cdot 100 = 235{,}5\ \text{cm}^2$. b) Radiuslar har xil, bo'laklar alohida: $\frac{1}{4} \cdot 3{,}14 \cdot 36 = 28{,}26\ \text{m}^2$ va $\frac{1}{4} \cdot 3{,}14 \cdot 9 = 7{,}065\ \text{m}^2$; $28{,}26 + 7{,}065 = 35{,}325\ \text{m}^2$.

8-masala

Har birining radiusi $4 : 2 = 2$ cm: $\frac{1}{2} \cdot 3{,}14 \cdot 2 \cdot 2 = 6{,}28\ \text{cm}^2$. Uchtasi: $3 \cdot 6{,}28 = 18{,}84\ \text{cm}^2$.

9-masala

Katta: $r = 8$ cm, $\frac{1}{2} \cdot 3{,}14 \cdot 64 = 100{,}48\ \text{cm}^2$. Kichiklarining diametri $16 : 2 = 8$ cm, radiusi 4 cm. Ikkalasi: $2 \cdot \frac{1}{2} \cdot 3{,}14 \cdot 16 = 50{,}24\ \text{cm}^2$. Bo'yalgan soha: $100{,}48 - 50{,}24 = 50{,}24\ \text{cm}^2$.

10-masala

Kesib olingan va yopishtirilgan yarim doiralar bir xil, shakl yuzi katta yarim doiraning yuziga teng: $r = 9$ cm, $\frac{1}{2} \cdot 3{,}14 \cdot 81 = 127{,}17\ \text{cm}^2$.

11-masala

$\frac{1}{4} \cdot 3{,}14 \cdot 16 = 12{,}56\ \text{m}^2$. Plitkalar: $12{,}56 \cdot 25 = 314$ ta.

12-masala

Har bir bo'lak — chorak doira: $\frac{1}{4} \cdot 3{,}14 \cdot 16 \cdot 16 = 200{,}96\ \text{cm}^2$.

Mustaqil ish

1-masala

$r = 13$ cm. $\frac{1}{2} \cdot 3{,}14 \cdot 169 = 265{,}33\ \text{cm}^2$.

2-masala

$\frac{1}{4} \cdot 22 \cdot 0{,}3 \cdot 2{,}1 = 3{,}465\ \text{m}^2$.

3-masala

$r = 3{,}7$ cm. $\pi \times 3{,}7 \times 3{,}7 \div 2 = 21{,}504\ldots \approx 21{,}5\ \text{cm}^2$.

4-masala

$\frac{3}{4} \cdot \frac{22}{7} \cdot 2{,}8 \cdot 2{,}8 = \frac{3}{4} \cdot 22 \cdot 0{,}4 \cdot 2{,}8 = 18{,}48\ \text{dm}^2$.

5-masala

Yarim doira, $r = 5$ cm: $\frac{1}{2} \cdot 3{,}14 \cdot 25 = 39{,}25\ \text{cm}^2$. Chorak doira: $\frac{1}{4} \cdot 3{,}14 \cdot 25 = 19{,}625\ \text{cm}^2$. Jami: $39{,}25 + 19{,}625 = 58{,}875\ \text{cm}^2$.

6-masala

Katta: $\frac{1}{2} \cdot 3{,}14 \cdot 36 = 56{,}52\ \text{cm}^2$. Kichik, $r = 3$ cm: $\frac{1}{2} \cdot 3{,}14 \cdot 9 = 14{,}13\ \text{cm}^2$. Qolgan qism: $56{,}52 - 14{,}13 = 42{,}39\ \text{cm}^2$.

7-masala

$r = 3$ m. $\frac{1}{2} \cdot 3{,}14 \cdot 9 = 14{,}13\ \text{m}^2$. Bo'yoq: $14{,}13 \cdot 150 = 2119{,}5$ g. 2 kg = 2000 g — yetmaydi, $2119{,}5 - 2000 = 119{,}5$ g kam.

8-masala

Nodira $\frac{1}{4}$ o'rniga $\frac{1}{2}$ ni oldi — bu yarim doiraning yuzi. To'g'risi: $\frac{1}{4} \cdot 3{,}14 \cdot 12 \cdot 12 = 113{,}04\ \text{cm}^2$.

9-masala

a) Chorak doira: $\frac{1}{4} \cdot 3{,}14 \cdot 100 = 78{,}5\ \text{cm}^2$. Yarim doira, $r = 5$ cm: $\frac{1}{2} \cdot 3{,}14 \cdot 25 = 39{,}25\ \text{cm}^2$. b) Chorak doiraniki katta: $78{,}5 : 39{,}25 = 2$ marta.